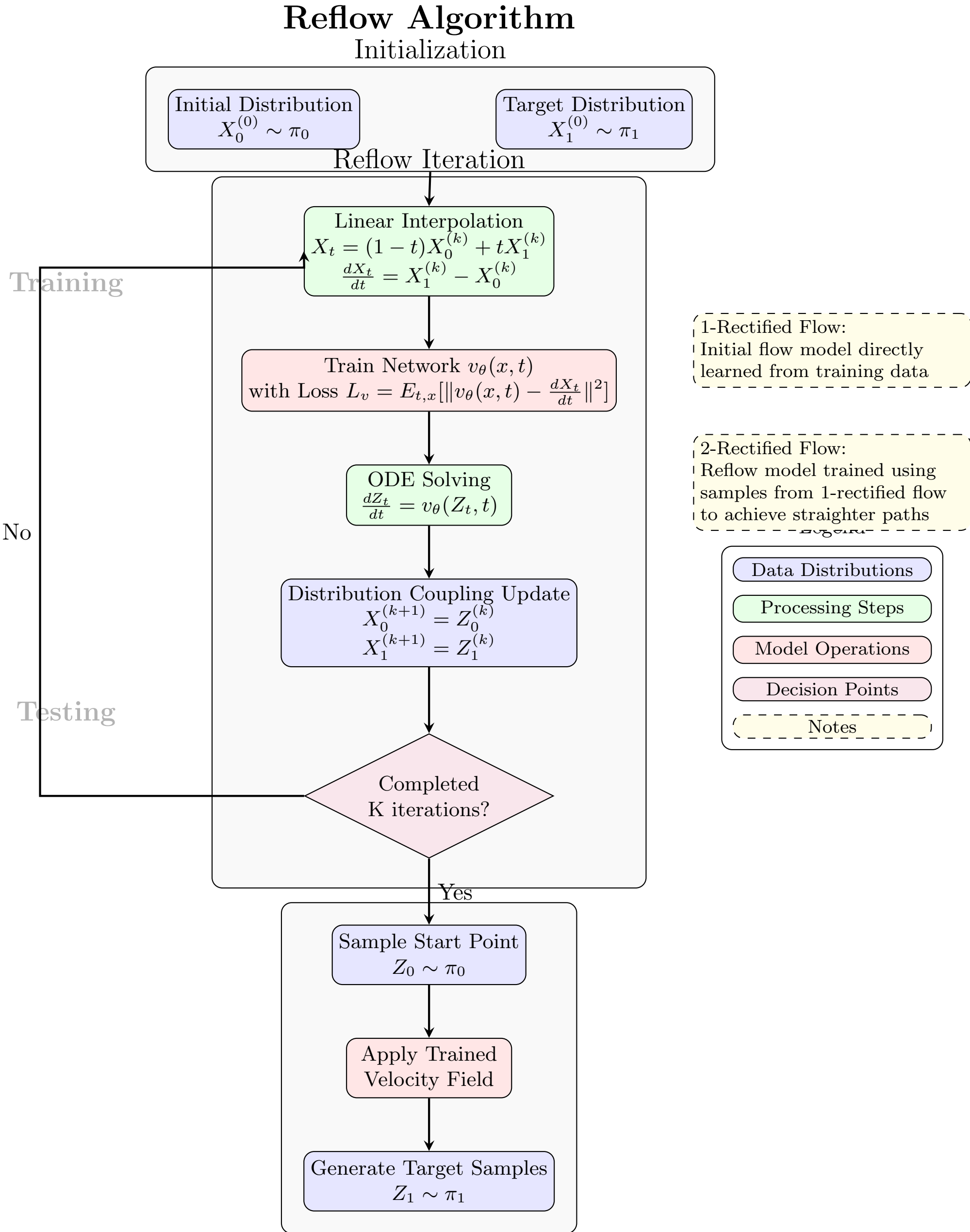
Reflow Algorithm
Initialization
Initial Distribution $X_0^{(0)} \sim \pi_0$
Target Distribution $X_1^{(0)} \sim \pi_1$
Reflow Iteration
Linear Interpolation
$X_t = (1-t)X_0^{(k)} + tX_1^{(k)}$
$\frac{dX_t}{dt} = X_1^{(k)} - X_0^{(k)}$
Training
Train Network $v_\theta(x,t)$
with Loss $L_v = E_{t,x}[\|v_\theta(x,t) - \frac{dX_t}{dt}\|^2]$
ODE Solving
$\frac{dZ_t}{dt} = v_\theta(Z_t, t)$
No
Distribution Coupling Update
$X_0^{(k+1)} = Z_0^{(k)}$
$X_1^{(k+1)} = Z_1^{(k)}$
Testing
Completed K iterations?
Yes
Sample Start Point $Z_0 \sim \pi_0$
Apply Trained Velocity Field
Generate Target Samples $Z_1 \sim \pi_1$
1-Rectified Flow: Initial flow model directly learned from training data
2-Rectified Flow: Reflow model trained using samples from 1-rectified flow to achieve straighter paths
Data Distributions
Processing Steps
Model Operations
Decision Points
Notes